

A censored gamma family for generalized additive models

Likelihood, derivatives and prediction targets

Implementation note for `mgcvUtils`

Prepared with Codex; draft for mathematical review

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This note accompanies `R/cgamma.R` on the `codex/cgamma` branch. It follows the separation between fitting and prediction in Miller’s censored lognormal note [1]. The implemented family has one mean predictor, a log link, and a single global dispersion parameter. It is not a gamma location–scale model. All logarithms below are natural.

1 Model and notation

For independent responses, conditional on the model coefficients,

$$Y_i \mid \boldsymbol{\beta}, \phi \sim \text{Gamma}\left(k, \text{rate} = \frac{k}{\mu_i}\right), \quad \eta_i = \mathbf{x}_i^\top \boldsymbol{\beta} + o_i, \quad \mu_i = e^{\eta_i}, \quad k = \phi^{-1}. \quad (1)$$

Thus $\mathbb{E}(Y_i \mid \boldsymbol{\beta}, \phi) = \mu_i$ and $\text{Var}(Y_i \mid \boldsymbol{\beta}, \phi) = \phi \mu_i^2$. Here o_i is a known offset. Internally the estimated family parameter is

$$\theta = \log \phi, \quad k = e^{-\theta}, \quad \partial_\theta k = -k, \quad \partial_\theta^2 k = k. \quad (2)$$

The public argument `theta` specifies dispersion on its original scale; `getTheta()` returns θ , whereas `getTheta(TRUE)` returns ϕ . This distinction is essential when checking derivatives.

Write $f(y; \mu, k)$ for the response density and $p_k(x)$ for the unit-rate gamma density. Define the regularised tails

$$P(k, x) = \frac{1}{\Gamma(k)} \int_0^x t^{k-1} e^{-t} dt, \quad Q(k, x) = 1 - P(k, x).$$

For a fixed response bound b , use $x_b = kb/\mu$. In particular,

$$\partial_\mu x_b = -x_b/\mu, \quad \partial_\theta x_b = -x_b, \quad \partial_\theta^2 x_b = x_b.$$

Partial derivatives in μ hold θ fixed, and vice versa. The digamma and trigamma functions are $\psi(k)$ and $\psi_1(k)$.

Prior weights $w_i \geq 0$ multiply log likelihoods: $\ell = \sum_i w_i \ell_i$. They are replication weights, not precision weights changing k_i . The additional `mgcv` scale is fixed at one because ϕ is already estimated as a family parameter.

2 Observed likelihood and saturated likelihood

Drop the observation subscript until needed. Censoring is assumed non-informative conditional on the covariates. With fixed censoring limits,

$$\ell(\mu, \theta) = \begin{cases} \log f(y; \mu, k), & \text{exact } y > 0, \\ \log P(k, kc/\mu), & Y \leq c, \\ \log Q(k, kc/\mu), & Y \geq c, \\ \log\{P(k, ku/\mu) - P(k, kl/\mu)\}, & l \leq Y \leq u. \end{cases} \quad (3)$$

For a continuous response, endpoint inclusion does not affect probabilities. One-sided censoring is an interval with $l = 0$ or $u = \infty$. Denote its probability by $A(\mu, \theta)$.

2.1 A common deviance definition

For each observation let

$$\ell_s(\theta) = \sup_{\mu > 0} \ell(\mu, \theta), \quad D(\mu, \theta) = 2w\{\ell_s(\theta) - \ell(\mu, \theta)\}.$$

For exact data $\mu_s = y$. For a proper one-sided censoring event the supremum probability is one, so $\ell_s = 0$. This also holds for $[0, u]$, $[l, \infty)$ and $[0, \infty)$; the supremum need not be attained at a finite positive mean.

For a finite interval $0 < l < u < \infty$, define

$$G(x, k) = xp_k(x) = \frac{x^k e^{-x}}{\Gamma(k)}. \quad (4)$$

Differentiating the probability gives

$$A_\mu = \frac{G(x_l, k) - G(x_u, k)}{\mu}. \quad (5)$$

The stationary equation is therefore

$$0 = \log G(x_l, k) - \log G(x_u, k) = k \log(l/u) + k(u-l)/\mu.$$

It has the unique solution

$$\boxed{\mu_s = \frac{u-l}{\log(u/l)}}. \quad (6)$$

The derivative is positive below μ_s and negative above it; interval probability tends to zero at either end of the positive mean domain. Thus this is the maximum. Crucially, μ_s is independent of θ :

$$\ell_s^{(j)}(\theta) = \partial_\theta^j \log A(\mu_s, \theta), \quad j = 1, 2.$$

No derivative of an optimised mean is needed. Numerically evaluate $\log(u/l)$ as $\log\{1 + (u-l)/l\}$ using `log1p`.

The identity used to check likelihood normalisation is

$$\frac{1}{2} \sum_i D_i - \sum_i w_i \ell_{s,i} = - \sum_i w_i \ell_i. \quad (7)$$

3 Exact observations: all required derivatives

Put $t = y/\mu$ and $x = kt$. The exact log density is

$$\ell = k \log k - k \log \mu - \log \Gamma(k) + (k-1) \log y - ky/\mu. \quad (8)$$

Since $\partial_\mu t = -t/\mu$, repeated differentiation gives

$$\ell_\mu = k(t-1)/\mu, \quad \ell_{\mu\mu} = k(1-2t)/\mu^2, \quad (9)$$

$$\ell_{\mu\mu\mu} = 2k(3t-1)/\mu^3, \quad \ell_{\mu\mu\mu\mu} = 6k(1-4t)/\mu^4. \quad (10)$$

Equivalently, for $r \geq 1$,

$$\ell_{\mu^r} = (-1)^r (r-1)! k(1-rt) \mu^{-r}. \quad (11)$$

Because t is independent of θ , each expression is proportional to $k = e^{-\theta}$, and hence

$$\boxed{\ell_{\mu^r\theta} = -\ell_{\mu^r}, \quad \ell_{\mu^r\theta\theta} = \ell_{\mu^r}.} \quad (12)$$

Only $r \leq 3$ is needed with one dispersion derivative, and $r \leq 2$ with two.

3.1 Pure dispersion derivatives

First differentiate (8) in k :

$$\ell_k = \log k + 1 - \psi(k) + \log t - t, \quad \ell_{kk} = 1/k - \psi_1(k).$$

Applying $\partial_\theta = -k\partial_k$ yields

$$s := \ell_\theta = -k\{\log k + 1 - \psi(k) + \log t - t\}, \quad (13)$$

$$\ell_{\theta\theta} = k\ell_k + k^2\ell_{kk} = -s + k - k^2\psi_1(k). \quad (14)$$

Evaluate these at $t = 1$ to obtain exact-observation saturated derivatives. As a useful consistency check,

$$D = 2wk\{t - 1 - \log t\}$$

is the usual gamma deviance divided by dispersion. Consequently $D_\theta = -D$ and $D_{\theta\theta} = D$ for exact observations.

3.2 Independent integral interpretation

For an interval with fixed bounds, differentiation under the integral gives

$$A_\theta = \int_l^u f(y)s(y)dy, \quad A_{\theta\theta} = \int_l^u f(y)\{s(y)^2 + \ell_{\theta\theta}(y)\}dy. \quad (15)$$

Thus $(\log A)_\theta = \mathbb{E}[s(Y) \mid l \leq Y \leq u]$ and $(\log A)_{\theta\theta} = \mathbb{E}[\ell_{\theta\theta}(Y) \mid l \leq Y \leq u] + \text{Var}[s(Y) \mid l \leq Y \leq u]$. These identities provide an independent quadrature check on the tail-derivative algorithm.

4 Censored observations: mean derivatives at the boundaries

From (4), holding k fixed,

$$\partial_\mu \log G(x_b, k) = (x_b - k)/\mu.$$

For any polynomial $B(x, k)$ and integer $r \geq 1$,

$$\partial_\mu \{\mu^{-r} G(x, k) B(x, k)\} = \mu^{-r-1} G(x, k) \{(x - k - r)B - xB_x\}. \quad (16)$$

Start from (5) and set $B_0 = 1$. Induction gives

$$B_r(x, k) = (x - k - r)B_{r-1}(x, k) - x\partial_x B_{r-1}(x, k), \quad (17)$$

$$A_{\mu^r} = \mu^{-r} \{G(x_l, k)B_{r-1}(x_l, k) - G(x_u, k)B_{r-1}(x_u, k)\}. \quad (18)$$

The necessary polynomials are

$$B_0 = 1, \quad (19)$$

$$B_1 = x - k - 1, \quad (20)$$

$$B_2 = x^2 - 2(k+2)x + (k+1)(k+2), \quad (21)$$

$$B_3 = x^3 - 3(k+3)x^2 + 3(k+2)(k+3)x - (k+1)(k+2)(k+3). \quad (22)$$

For $k > 0$, the boundary terms, and their required derivatives, vanish as $x \downarrow 0$ or $x \rightarrow \infty$. In computation they are set to zero at these endpoints rather than evaluating $0 \log 0$ or $\infty - \infty$.

4.1 Convert probability derivatives to log-probability derivatives

Let $R_r = A_{\mu^r}/A$, and write $L_r = \partial_\mu^r \log A$. Repeated application of the quotient rule gives

$$L_1 = R_1, \quad (23)$$

$$L_2 = R_2 - R_1^2, \quad (24)$$

$$L_3 = R_3 - 3R_1R_2 + 2R_1^3, \quad (25)$$

$$L_4 = R_4 - 4R_1R_3 - 3R_2^2 + 12R_1^2R_2 - 6R_1^4. \quad (26)$$

For example, $\partial_\mu R_r = R_{r+1} - R_r R_1$, which verifies each line from the previous one.

Compute the normalised boundary terms as

$$\frac{G(x_b, k)}{A} = \exp\{\log G(x_b, k) - \log A\}.$$

This avoids forming a potentially underflowed probability A explicitly. The same expressions cover left, right and finite-interval censoring; only the boundaries change. Narrow finite intervals require the alternative in Section 8 to avoid subtracting nearly identical boundary terms.

5 Mixed derivatives through second-order forward differentiation

Represent a scalar function of θ by the triple

$$\mathcal{J}(a) = (a, \dot{a}, \ddot{a}), \quad \dot{a} = \partial_\theta a, \quad \ddot{a} = \partial_\theta^2 a.$$

These are derivatives, not Taylor coefficients. Addition and subtraction are componentwise; constants have zero derivative components. Product, reciprocal, logarithm and exponential rules are

$$\mathcal{J}(ab) = (ab, \dot{a}b + a\dot{b}, \ddot{a}b + 2\dot{a}\dot{b} + a\ddot{b}), \quad (27)$$

$$\mathcal{J}(a^{-1}) = (a^{-1}, -\dot{a}/a^2, 2\dot{a}^2/a^3 - \ddot{a}/a^2), \quad (28)$$

$$\mathcal{J}(\log a) = (\log a, \dot{a}/a, \ddot{a}/a - (\dot{a}/a)^2), \quad (29)$$

$$\mathcal{J}(e^a) = (e^a, \dot{a}e^a, \ddot{a}e^a + \dot{a}^2e^a). \quad (30)$$

All quotient operations below mean multiplication by the reciprocal triple.

5.1 Boundary triples

For $x = kb/\mu$, initialise $\mathcal{J}(k) = (k, -k, k)$ and $\mathcal{J}(x) = (x, -x, x)$. Writing $q = \log G = k \log x - x - \log \Gamma(k)$ gives

$$\dot{q} = x - k\{\log x + 1 - \psi(k)\}, \quad (31)$$

$$\ddot{q} = -\dot{q} + k - k^2\psi_1(k). \quad (32)$$

Here both x and k vary with θ . Treating the argument of the gamma CDF as fixed when differentiating shape would omit part of the derivative.

Given $\mathcal{J}(\log A)$ from Sections 6–7, form $\mathcal{J}(G/A) = \mathcal{J}[\exp(q - \log A)]$. Evaluate the polynomials in (22) using triples, construct $\mathcal{J}(R_r)$ using (18), and apply (26) using the product rule above. The output $\mathcal{J}(L_r)$ contains ℓ_{μ^r} , $\ell_{\mu^r\theta}$ and $\ell_{\mu^r\theta\theta}$.

5.2 Explicit mixed expressions for checking

The first two orders, written without triple notation, are

$$\dot{L}_1 = \dot{R}_1, \quad \ddot{L}_1 = \ddot{R}_1, \quad (33)$$

$$\dot{L}_2 = \dot{R}_2 - 2R_1\dot{R}_1, \quad \ddot{L}_2 = \ddot{R}_2 - 2\{\dot{R}_1^2 + R_1\ddot{R}_1\}, \quad (34)$$

$$\dot{L}_3 = \dot{R}_3 - 3\{\dot{R}_1R_2 + R_1\dot{R}_2\} + 6R_1^2\dot{R}_1.$$

Only these mixed expressions are required by Dd; L_4 itself needs no dispersion derivative. The implementation carries a complete triple through the polynomial arithmetic for simplicity. No finite difference is used here.

6 Dispersion derivatives of gamma tails: lower-tail series

The incomplete-gamma expansion in [2] can be written

$$P(k, x) = \frac{e^{q(x, k)}}{k} \sum_{j=0}^{\infty} T_j, \quad T_0 = 1, \quad T_j = T_{j-1} \frac{x}{k+j}. \quad (35)$$

The implementation uses this expansion for $x < k+1$. All terms are positive. At fixed observation bound and mean, $\dot{x} = -x$ and $\dot{k} = -k$. For the ratio $r_j = x/(k+j)$,

$$\partial_{\theta} \log r_j = -1 + \frac{k}{k+j} = -\frac{j}{k+j}, \quad \partial_{\theta}^2 \log r_j = -\frac{jk}{(k+j)^2}.$$

Consequently an explicit derivative recurrence is

$$\dot{r}_j = -r_j \frac{j}{k+j}, \quad (36)$$

$$\ddot{r}_j = r_j \frac{j(j-k)}{(k+j)^2}, \quad (37)$$

$$\dot{T}_j = \dot{T}_{j-1} r_j + T_{j-1} \dot{r}_j, \quad (38)$$

$$\ddot{T}_j = \ddot{T}_{j-1} r_j + 2\dot{T}_{j-1} \dot{r}_j + T_{j-1} \ddot{r}_j. \quad (39)$$

Initialise $\dot{T}_0 = \ddot{T}_0 = 0$. Sum these sequences to obtain $S = \sum T_j$, $\dot{S} = \sum \dot{T}_j$, $\ddot{S} = \sum \ddot{T}_j$. Taking the logarithm in (35) gives

$$\log P = q - \log k + \log S, \quad (40)$$

$$\partial_{\theta} \log P = \dot{q} + 1 + \dot{S}/S, \quad (41)$$

$$\partial_{\theta}^2 \log P = \ddot{q} + \ddot{S}/S - (\dot{S}/S)^2. \quad (42)$$

The $+1$ in the first derivative is from $-\log k = \theta$; there is no corresponding second-derivative term.

6.1 Implementation and convergence

The code computes the recurrence with the rules in (30) rather than maintaining separate hand-expanded recurrences. It stops when all three components of the new term satisfy

$$|\mathcal{J}(T_j)_m| < 2 \times 10^{-14} \{1 + |\mathcal{J}(S)_m|\}, \quad m = 0, 1, 2.$$

There is a hard limit of 10,000 iterations with an error on nonconvergence. Convergence of the probability value alone is not sufficient for convergence of its derivatives.

R's `pgamma(log.p = TRUE)` supplies the final log-probability value. The differentiated series supplies its first two dispersion derivatives. This preserves R's specialised tail evaluation without numerically perturbing the dispersion. Very large shapes near the transition region remain a regime to test beyond the present finite validation grid.

7 Dispersion derivatives of gamma tails: upper-tail fraction

An equivalent contracted form of the upper incomplete-gamma continued fraction [3] is

$$Q(k, x) = e^{q(x, k)} H(k, x), \quad H = \frac{1}{b_0 + \frac{a_1}{b_1 + \frac{a_2}{b_2 + \dots}}}, \quad b_j = x + 1 - k + 2j, \quad a_j = j(k - j). \quad (43)$$

Use this representation for $x \geq k + 1$. The triple inputs for its coefficients are particularly simple:

$$\mathcal{J}(b_j) = (b_j, -x + k, x - k), \quad \mathcal{J}(a_j) = (a_j, -jk, jk). \quad (44)$$

7.1 Differentiating the fraction evaluation

A multiplicative evaluation starts with

$$C_0 = M, \quad D_0 = b_0^{-1}, \quad H_0 = D_0, \quad M = 10^{300},$$

where M approximates an infinite initial C_0 and its derivative components are zero. For $j = 1, 2, \dots$ compute

$$D_j = (b_j + a_j D_{j-1})^{-1}, \quad (45)$$

$$C_j = b_j + a_j / C_{j-1}, \quad (46)$$

$$\Delta_j = C_j D_j, \quad (47)$$

$$H_j = H_{j-1} \Delta_j. \quad (48)$$

Replace each operation by its triple rule in (30). This yields H , $\dot{H}$ and $\ddot{H}$, and hence

$$\log Q = q + \log H, \quad (49)$$

$$\partial_\theta \log Q = \dot{q} + \dot{H}/H, \quad (50)$$

$$\partial_\theta^2 \log Q = \ddot{q} + \ddot{H}/H - (\dot{H}/H)^2. \quad (51)$$

The stopping rule checks $|\Delta_j - 1|$, $|\dot{\Delta}_j|$ and $|\ddot{\Delta}_j|$ against 2×10^{-14} , with the same iteration limit as the series. Derivatives of a fraction that terminates in value at integer k can still require further iterations: the derivative tests must remain.

The upper-tail value is again supplied by `pgamma`, with `lower.tail = FALSE`. Derivatives are obtained from the differentiated fraction. A requested tail opposite to the directly evaluated tail is obtained by the log-difference rule on the next page.

These expressions specify a numerical evaluation of analytic derivatives. They are not finite differences, but finite iteration and floating-point arithmetic still require numerical validation. No assertion of uniform accuracy over all positive shapes is made by the present implementation.

8 Stable interval probabilities and narrow intervals

Let $a = \log A_1$, $b = \log A_0$ with $A_1 > A_0 \geq 0$ and define $z = \log(A_1 - A_0)$. Evaluate

$$z = a + \log\{-\expm1(b - a)\}, \quad r = e^{b-z}, \quad d = \dot{a} - \dot{b}. \quad (52)$$

Direct differentiation gives

$$\dot{z} = \dot{a} + rd, \quad (53)$$

$$\ddot{z} = \ddot{a} + r(\ddot{a} - \ddot{b}) - r(1 + r)d^2. \quad (54)$$

This also computes a complement by setting $a = 0$ with zero derivatives. At a tail endpoint having probability zero, use log value $-\infty$ and zero derivative components, so $r = 0$.

For a finite interval either subtract lower tails, $P(k, x_u) - P(k, x_l)$, or upper tails, $Q(k, x_l) - Q(k, x_u)$. The current implementation uses lower tails if $P(k, x_l) < 1/2$, otherwise upper tails. This avoids subtracting two near-one probabilities in the far upper tail, although it cannot remove all cancellation for narrow intervals.

8.1 Integrate the density and its derivatives together

For $\delta = u - l$ small, substitute $y = l + \delta(v + 1)/2$:

$$A = \frac{\delta}{2} \int_{-1}^1 f\{l + \delta(v + 1)/2\} dv \simeq \frac{\delta}{2} \sum_{j=1}^{16} \omega_j f(y_j). \quad (55)$$

The code uses 16-point Gauss–Legendre quadrature when $0 < l < u < \infty$, $\delta/l < 10^{-4}$ and $k\delta/\mu < 1$. This is a local numerical approximation, not a replacement likelihood model.

For density derivatives define $C_0 = 1$ and

$$\partial_\mu^r f = f \mu^{-r} C_r(x, k), \quad C_{r+1} = (x - k - r)C_r - x \partial_x C_r.$$

Thus $C_1 = x - k$, $C_2 = x^2 - 2(k + 1)x + k(k + 1)$, and

$$\begin{aligned} C_3 &= x^3 - 3(k + 2)x^2 + 3(k + 1)(k + 2)x - k(k + 1)(k + 2), \\ C_4 &= x^4 - 4(k + 3)x^3 + 6(k + 2)(k + 3)x^2 \\ &\quad - 4(k + 1)(k + 2)(k + 3)x + k(k + 1)(k + 2)(k + 3). \end{aligned}$$

Integrate $f C_r / \mu^r$ with the same nodes and weights, divide by the integral of f , and use (26). Carry all operations in θ triples. A common subtraction of the largest node log density prevents underflow; treat this numerical scaling constant as fixed within the derivative calculation. As $\delta \downarrow 0$, log probability approaches $\log f\{(l + u)/2\} + \log \delta$ and its parameter derivatives approach those of an exact observation. This is a separate regression test.

9 Avoiding cancellation in far upper-tail derivatives

The boundary formula is algebraically correct but can be numerically poor far into the upper tail. For example, R_4 and R_1^4 can both have order x^4 while their combination in L_4 has order x . Subtracting these large quantities can destroy the fourth derivative. Working on the log probability scale alone does not resolve this cancellation.

Repeated integration by parts gives the upper-tail expansion

$$Q(k, x) \sim \frac{G(x, k)}{x} S(x, k), \quad S = \sum_{j \geq 0} U_j, \quad U_0 = 1, \quad U_j = U_{j-1} \frac{k-j}{x}. \quad (56)$$

This is an asymptotic expansion; it must be truncated while the terms are small, not summed indefinitely. The code uses it only for $x > \max(100, 5k)$ and requires convergence of all retained derivatives. Since $x = kb/\mu$, U_j is proportional to μ^j when k is fixed:

$$S_{\mu^r} = \mu^{-r} \sum_{j \geq r} j(j-1) \cdots (j-r+1) U_j, \quad r = 1, \dots, 4.$$

Evaluate these sums as θ triples, normalise by S , and use (26) to get derivatives of $\log S$. Then

$$(\log Q)_{\mu^r} = (-1)^r (r-1)! \mu^{-r} \{k-1-rx\} + (\log S)_{\mu^r}. \quad (57)$$

The dominant order- x term is now explicit, and the remaining small series does not require cancellation of order- x^4 terms. Dispersion and mixed derivatives follow by carrying the same triples throughout.

9.1 Finite intervals wholly in the far upper tail

Write $L = \log Q(k, x_l)$, $U = \log Q(k, x_u)$, $\Delta = U - L$, $e = \exp \Delta$ and $F = 1 - e$. Then $\log A = L + \log F$. For $\Delta_r = \partial_{\mu}^r \Delta$,

$$\begin{aligned} e_{\mu} &= e \Delta_1, \\ e_{\mu^2} &= e(\Delta_2 + \Delta_1^2), \\ e_{\mu^3} &= e(\Delta_3 + 3\Delta_1 \Delta_2 + \Delta_1^3), \\ e_{\mu^4} &= e(\Delta_4 + 4\Delta_1 \Delta_3 + 3\Delta_2^2 + 6\Delta_1^2 \Delta_2 + \Delta_1^4). \end{aligned}$$

Because $F_{\mu^r} = -e_{\mu^r}$, applying (26) to the ratios $-e_{\mu^r}/F$ supplies $(\log F)_{\mu^r}$. Add these to the derivatives of L from (57). This calculation also uses triples for mixed derivatives. The narrow-interval quadrature takes precedence when its conditions hold. For $u = \infty$, $\log A = L$ directly.

10 Mapping the derivatives to the `mgcv` interface

The extended-family interface requires derivatives of deviance in μ , not derivatives in η . For $r \geq 1$ the saturated likelihood does not depend on the fitted mean, giving

$$D_{\mu^r \theta^s} = -2w\ell_{\mu^r \theta^s}, \quad D_\theta = 2w(\ell_{s,\theta} - \ell_\theta), \quad D_{\theta\theta} = 2w(\ell_{s,\theta\theta} - \ell_{\theta\theta}).$$

The return fields of `Dd(y, mu, theta, wt, level)` are:

Level	Field	Derivative
0	<code>Dmu</code> , <code>Dmu2</code>	D_μ , $D_{\mu\mu}$
1	<code>Dmu3</code> , <code>Dth</code>	$D_{\mu\mu\mu}$, D_θ
1	<code>Dmuth</code> , <code>Dmu2th</code>	$D_{\mu\theta}$, $D_{\mu\mu\theta}$
2	<code>Dmu4</code> , <code>Dth2</code>	$D_{\mu\mu\mu\mu}$, $D_{\theta\theta}$
2	<code>Dmuth2</code> , <code>Dmu2th2</code>	$D_{\mu\theta\theta}$, $D_{\mu\mu\theta\theta}$
2	<code>Dmu3th</code>	$D_{\mu\mu\mu\theta}$

Levels are cumulative. For exact data use the Gamma expected information: `EDmu2` = $2wk/\mu^2$ and `EDmu2th` = $-2wk/\mu^2$. For censored data use the working approximation `EDmu2` = $\max(0, D_{\mu\mu} + D_\mu/\mu)$ and, where positive, `EDmu2th` = $D_{\mu\mu\theta} + D_{\mu\theta}/\mu$ (zero otherwise). The latter uses non-negative observed curvature in η , not exact expected information. Using $D_{\mu\mu}$ alone can give negative working weights in `bam`, even when $D_{\eta\eta} > 0$. Only the exact-data choice is required to recover Gamma's EDF and covariance conventions.

The `ls` function returns $\sum_i w_i \ell_{s,i}$, its first two θ derivatives, and the per-observation first derivatives as `LSTH1`. The `aic` function returns $-2 \sum_i w_i \ell_i$; `mgcv` adds the model complexity contribution. Zero weights contribute zero to all of these quantities.

10.1 Log-link conversion and checks

Since $\partial_\eta = \mu \partial_\mu$,

$$D_\eta = \mu D_\mu, \tag{58}$$

$$D_{\eta\eta} = \mu^2 D_{\mu\mu} + \mu D_\mu, \tag{59}$$

$$D_{\eta^3} = \mu^3 D_{\mu^3} + 3\mu^2 D_{\mu^2} + \mu D_\mu, \tag{60}$$

$$D_{\eta^4} = \mu^4 D_{\mu^4} + 6\mu^3 D_{\mu^3} + 7\mu^2 D_{\mu^2} + \mu D_\mu. \tag{61}$$

The same linear combinations hold after a θ derivative because μ is fixed in that derivative. The family supplies the μ derivatives; `mgcv` performs this conversion.

For an exact observation, direct differentiation of $\ell = \text{constant} - k\eta - kye^{-\eta}$ gives $\ell_\eta = x - k$, $\ell_{\eta^2} = -x$, $\ell_{\eta^3} = x$ and $\ell_{\eta^4} = -x$. These independently check (61). For uncensored data the fitted model matches an ordinary Gamma fit under ML or REML, including likelihood dispersion, coefficients and EDF. The extended-family smoothing parameter equals the Gamma smoothing parameter divided by ϕ . Gamma stores its likelihood dispersion in `reml.scale`; `sig2` and `scale` use a separate Fletcher estimate by default. To compare covariance matrices and standard errors, rescale the ordinary Gamma covariance by `reml.scale / sig2`. The censored-family external scale stays at one, with dispersion in `family$getTheta(TRUE)`.

11 Prediction: conditional mean and epsilon correction

The implemented response prediction is

$$\hat{\mu}_0 = g(\hat{\beta}) = \exp(\mathbf{x}_0^\top \hat{\beta} + o_0).$$

This is already a plug-in estimate of the conditional arithmetic mean. There is no lognormal residual-variance adjustment. A different target is $\mathbb{E}\{g(\beta) \mid \mathbf{y}, \hat{\phi}, \hat{\lambda}\}$, which averages a nonlinear function over coefficient uncertainty. The epsilon approach of Thorson and Kristensen [4], based on Tierney, Kass and Kadane [5], is relevant to this target.

11.1 Expansion at the penalised mode

Fix dispersion and smoothing parameters for this derivation. Define the negative log posterior, up to a constant, by

$$h(\beta) = -\sum_i w_i \ell_i(\eta_i, \theta) + \frac{1}{2} \beta^\top S \beta.$$

At its mode let $V = \{h''(\hat{\beta})\}^{-1}$, $T = h'''(\hat{\beta})$ and $Q = h''''(\hat{\beta})$. Unlike Miller's per-observation h convention, h here is the full negative log posterior, so there are no explicit factors of sample size. The fully exponential Laplace expansion gives

$$\mathbb{E}(g \mid \mathbf{y}) \simeq g + \frac{1}{2} \text{tr}(V g'') - \frac{1}{2} T[V, V g']. \quad (62)$$

In components $T[V, V g'] = \sum_{abc} T_{abc} V_{ab} (V g')_c$. The last term corrects for departure from a Gaussian posterior and must not be dropped merely because a Gaussian covariance estimate is available.

For $g = e^{\mathbf{x}_0^\top \beta + o_0}$, $g' = g \mathbf{x}_0$ and $g'' = g \mathbf{x}_0 \mathbf{x}_0^\top$. Also

$$T = \sum_i t_i \mathbf{x}_i^{\otimes 3}, \quad Q = \sum_i q_i \mathbf{x}_i^{\otimes 4}, \quad t_i = \frac{1}{2} D_{i, \eta^3}, \quad q_i = \frac{1}{2} D_{i, \eta^4}.$$

The quadratic penalty has no third or fourth derivative. Set $v_0 = \mathbf{x}_0^\top V \mathbf{x}_0$, $d_i = \mathbf{x}_i^\top V \mathbf{x}_i$ and $e_i = \mathbf{x}_i^\top V \mathbf{x}_0$. Then (62) reduces to

$$\mathbb{E}(g \mid \mathbf{y}) \simeq g \left\{ 1 + \frac{1}{2} v_0 - \frac{1}{2} \sum_i t_i d_i e_i \right\}. \quad (63)$$

For exact gamma observations $t_i = -w_i k y_i / \mu_i$ and $q_i = w_i k y_i / \mu_i$. Therefore *all observations*, not just censored ones, contribute to the correction. This prevents direct reuse of the censored-row-only calculation for the lognormal family.

12 Checking the prediction expansion and its variance

12.1 An exact intercept-only check

Consider n uncensored observations, unit weights, fixed k , and a flat prior on $\eta = \log \mu$. Write $a = nk$ and $b = k \sum_i y_i$. The likelihood in η is proportional to $\exp(-a\eta - be^{-\eta})$. The change of variables $\mu = e^\eta$ shows

$$\mu \mid \mathbf{y} \sim \text{InverseGamma}(a, b), \quad \mathbb{E}(\mu \mid \mathbf{y}) = \frac{b}{a-1} \quad (a > 1).$$

At the mode in η , $g = b/a = \bar{y}$, $V = 1/a$, $T = -a$ and $Q = a$. Equation (62) therefore yields

$$g + \frac{g}{2a} + \frac{g}{2a} = g(1 + 1/a),$$

which agrees with the expansion of $b/(a-1)$ to this order. Retaining only the curvature term gives $g(1 + 1/(2a))$ and misses half of the leading correction. The exact posterior exists for $a > 0$, but its mean requires $a > 1$; a local expansion cannot guarantee existence of posterior moments.

12.2 Variance expression to be checked before implementation

For a scalar prediction define

$$\mathbf{a} = Vg', \quad B = g'' - T[\mathbf{a}], \quad \mathbf{b} = V\{2g''\mathbf{a} - T[\mathbf{a}, \mathbf{a}]\}.$$

Here $T[\mathbf{a}]$ is a matrix, $T[\mathbf{a}, \mathbf{a}]$ a vector, and $Q[\mathbf{a}, \mathbf{a}]$ a matrix. Differentiating the log determinant of the perturbed Hessian in the fully exponential approximation gives

$$\text{Var}(g \mid \mathbf{y}) \simeq g'^T V g' + \frac{1}{2} \text{tr}\{(VB)^2\} - \frac{1}{2} \text{tr}[V\{Q[\mathbf{a}, \mathbf{a}] + T[\mathbf{b}] - 2g'''[\mathbf{a}]\}]. \quad (64)$$

For the intercept-only check above, this equals $g^2\{a^{-1} + 4a^{-2}\}$, matching the expansion of the exact variance $b^2/\{(a-1)^2(a-2)\}$ for $a > 2$.

12.3 Scope of the prediction result

Equations (63) and (64) are derivations for a future prediction option, not a currently implemented prediction method. They condition on estimated dispersion and smoothing parameters. Joint integration over these quantities requires corresponding joint derivatives and covariance information. Prediction for a *new* random-effect level also introduces a different integral and is not supplied by these formulae alone.

The code presently follows ordinary `Gamma(link = "log")` prediction semantics. Posterior-mean prediction, frequentist bias correction of fixed parameters, and marginal prediction over new random effects should not be presented as interchangeable targets. Numerical integration of small models and simulation should precede exposing an epsilon-corrected option.

13 Verification and reviewer guide

The companion regression checks are in `tests/cgamma.R`. They cover:

1. First through fourth mean derivatives and every required mixed derivative, compared with Richardson differences of lower-order quantities. Exact, left-, right-, ordinary interval and narrow interval responses are included, together with support endpoints and zero weights.
2. Pure dispersion derivatives of log tails, compared with independently perturbed calls to base R's `pgamma`. The shape grid is 0.03, 0.2, 1, 2, 10, 100, 1000 and the bound-to-mean ratio grid is 0.001, 0.1, 0.9, 1, 1.1, 2, 10, 100, for both tails. Independent numerical integrals of the density score and its derivative also check dispersion derivatives for each censoring direction. Far upper-tail mean and mixed derivatives are checked separately.
3. Saturated-likelihood derivatives and the exact likelihood identity (7); gamma density and deviance formulae in the uncensored case.
4. The narrow-interval limit, including derivatives; an ordinary gamma smooth fit at fixed and estimated dispersion (ML and REML, vector and two-column exact responses); and a censored regression fit against independent direct likelihood optimisation.
5. Estimated-dispersion smooth fits, offsets, response predictions, preservation of censoring when subsetting, optimised null deviance, both `bam` fitting paths, and input validation.

These checks detect algebraic and implementation errors; they are not a proof of uniform accuracy or a large simulation study of estimator behaviour. Tail calculations at more extreme shapes, censoring-driven weak identifiability, and epsilon-corrected predictions deserve further study.

For review, the most consequential checks are the sign change $\theta = \log \phi = -\log k$; the simultaneous change of x and k in (31); the saturated interval mean in (6); the boundary recurrence (17); and the distinction between posterior mean and plug-in prediction in Section 11.

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